

Expense Tracker Project Documentation

1. Introduction

This project is a Java-based Expense Tracker application designed to help users record, manage, search, and categorize their daily expenses.

2. Objectives

- Apply basic OOP (Object-Oriented Programming) concepts in Java.
- Manage a list of expenses using ArrayList.
- Implement add, delete, search, and display functionalities.
- Provide category-wise and total expense calculations.

3. Features Implemented

- Add new expense
- Display all expenses
- Auto-generated Expense ID
- Search expense by keyword
- Delete expense by ID
- Total expense calculation
- Category-wise total expense

4. Tools & Technologies

Description	Tool Used
Programming Language	Java
Editor/IDE	VS Code / IntelliJ / Eclipse
Version Control	Git & GitHub

5. Project Structure

- Expense.java
- ExpenseTracker.java
- Main.java
- screenshots/
- README.md

6. Output Screenshots

```
Expense Tracker Menu
1. Add Expense
2. Show All Expenses
3. Show Total Expenses
4. Show By Category
5. Search Expense
6. Delete Expense
7. Exit
Enter your choice: 1
Enter amount: 210
Enter category: Food
Enter description: Lunch at canteen
Enter date: 21/10/2026
Expense Added!
```

```
Expense Tracker Menu
1. Add Expense
2. Show All Expenses
3. Show Total Expenses
4. Show By Category
5. Search Expense
6. Delete Expense
7. Exit
Enter your choice: 2
The ID is: 1
The expense is: 210.0
The category is: Food
The description is: Lunch at canteen
The date is: 21/10/2026
The ID is: 2
The expense is: 7800.0
The category is: Education
The description is: Fees
The date is: 12/09/2026
```

```
Expense Tracker Menu
1. Add Expense
2. Show All Expenses
3. Show Total Expenses
4. Show By Category
5. Search Expense
6. Delete Expense
7. Exit
Enter your choice: 3
Total expenses: 8010.0
```

```
Expense Tracker Menu
1. Add Expense
2. Show All Expenses
3. Show Total Expenses
4. Show By Category
5. Search Expense
6. Delete Expense
7. Exit
Enter your choice: 4
Enter the category: Education
The ID is: 2
The expense is: 7800.0
The category is: Education
The description is: Fees
The date is: 12/09/2026
```

```
Expense Tracker Menu
1. Add Expense
2. Show All Expenses
3. Show Total Expenses
4. Show By Category
5. Search Expense
6. Delete Expense
7. Exit
Enter your choice: 6
Enter ID to delete: 1
Deleted!
```

```
Expense Tracker Menu
1. Add Expense
2. Show All Expenses
3. Show Total Expenses
4. Show By Category
5. Search Expense
6. Delete Expense
7. Exit
Enter your choice: 7
Exiting
```

7. Conclusion

This project demonstrates the practical usage of Java OOP, arrays/lists, and console-based interaction. It fulfils the required guidelines for the flipped course project.